Monotone regression bands

Isotonic COSI constructions and a simulation comparison

28 October 2026

12 settings × 4 sample sizes × 500 Monte Carlo replications Nominal level: 95% COSI tuning: $\delta = 0.30$

This study compares two COSI confidence-band constructions based on isotonic estimators, an independent isotonic pilot, and block-count scales. It compares simultaneous containment and finite interior band width across smooth signals, flat regions, a jump, unequal local slopes, changing noise variance, and heavy tails. The shaded-band illustrations use the first replication at each sample size. Companion materials provide the R implementation, raw simulation records, diagnostics, and figures.

1 Data and simulation design

We generate independent observations $Y_i = f_0(X_i) + \xi_i$, $1 \le i \le n$, and compare six procedures for a monotone regression band. The sample sizes are $n \in \{100, 250, 500, 1000\}$, with 500 Monte Carlo replications for each of the twelve settings in Table 1. The nominal level is 95%. This report describes the implemented constructions and their empirical behavior. All reported coverage values are Monte Carlo frequencies.

Write $z = x - 1/2$, let $Z \sim N(0,1)$, and let T_ν have the Student t distribution with ν degrees of freedom. For the skewed-error setting, define

$$P_a = \sqrt{a(a-2)}\left\{\frac{a-1}{a}V_a - 1\right\}, \qquad \mathbb{P}(V_a > v) = v^{-a} \quad (v \ge 1), \qquad a = 2.5.$$

Thus P_a has mean zero and variance one. Set $c_{1.5} = \Phi^{-1}(3/4)/F_{t_{1.5}}^{-1}(3/4)$, so that $c_{1.5}T_{1.5}$ and Z have the same interquartile range. Unless the table specifies otherwise, $X \sim \mathrm{Unif}(0,1)$ and the error-generating variable is independent of X. In setting 11, S is Rademacher, $U \sim \mathrm{Unif}(0,1)$, and S, U, Z are independent. All error distributions have mean zero; setting 12 has infinite variance.

Setting	$f_0(x)$	Covariate distribution	Error ξ
1	x	Uniform	$0.2Z$
2	$(e^{2x}-1)/(e^2-1)$	Uniform	$0.2Z$
3	$1/2 + 4z^3$	Uniform	$0.2Z$
4	x	Beta$(2,3)$	$(0.1 + 0.4X)Z$
5	x	Uniform	$0.2T_3/\sqrt{3}$
6	x	Uniform	$0.2P_{2.5}$
7	$1/2$	Uniform	$0.2Z$
8	$4(z_+)^2$	Uniform	$0.2Z$
9	$\begin{cases} 0.2 + 0.4z, & z \le 0, \\ 0.2 + 1.6z, & z > 0. \end{cases}$	Uniform	$0.2Z$
10	$0.4x + 0.6\mathbf{1}\{x \ge 1/2\}$	Uniform	$0.2Z$
11	x	$1/2 + S\sqrt{U}/2$	$0.2Z$
12	x	Uniform	$0.2c_{1.5}T_{1.5}$

Table 1: The twelve monotone regression settings. Settings 7–12 probe flatness, unequal local slopes, a jump, a design density that vanishes linearly at the center, and errors without a second moment. Setting 8 has a flat region followed by a quadratic rise.

The local-shape distinctions are motivated by (Mallick et al., 2026, Section 2.3, especially Example 2.3; Section 3.2; Appendix A). We record the empirical frequencies of underestimation, overestimation, and equality separately at every anchor for the split, pilot, and full-sample estimators. We also record how often an anchor lies outside a subsample's observed covariate range.

1.1 One-replication illustrations

Figures 1–12 illustrate the bands for the prespecified first replication in every setting, at $n = 100, 250, 500, 1000$. The shaded region is cosi-m+ and the dashed red boundaries show HulC. The black curve is the truth, the blue step curve is the full-sample isotonic fit, and the gray points are the observations. Datasets are not nested across sample sizes. All band boundaries on $[0.1, 0.9]$ are displayed, although extreme responses can fall outside the vertical range. Companion figures give the corresponding cosi-m rectangular enclosures. The constructions and width measurements are described after the illustrations.

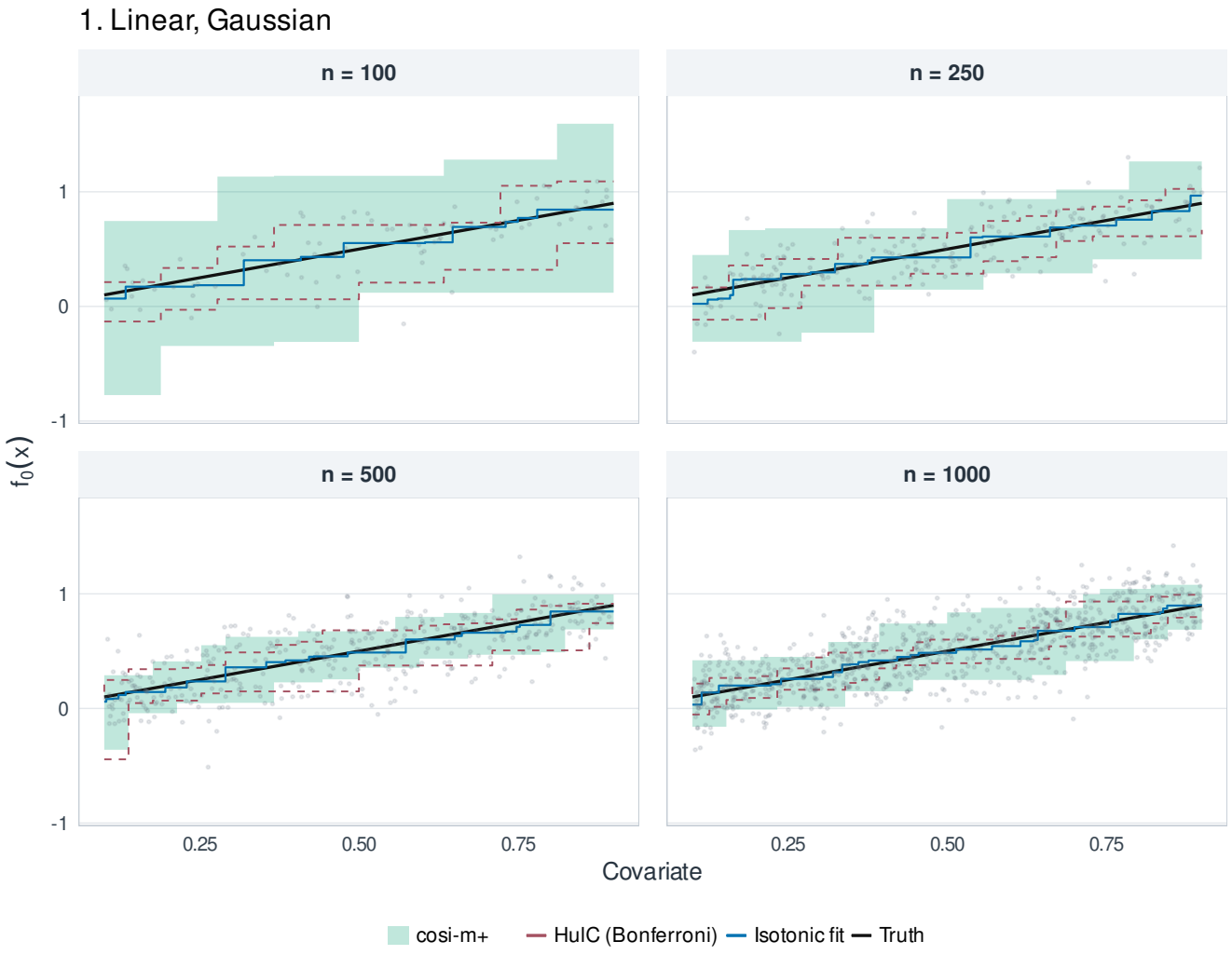


Figure 1: Setting 1: Linear mean, uniform covariates and Gaussian errors with standard deviation 0.2.

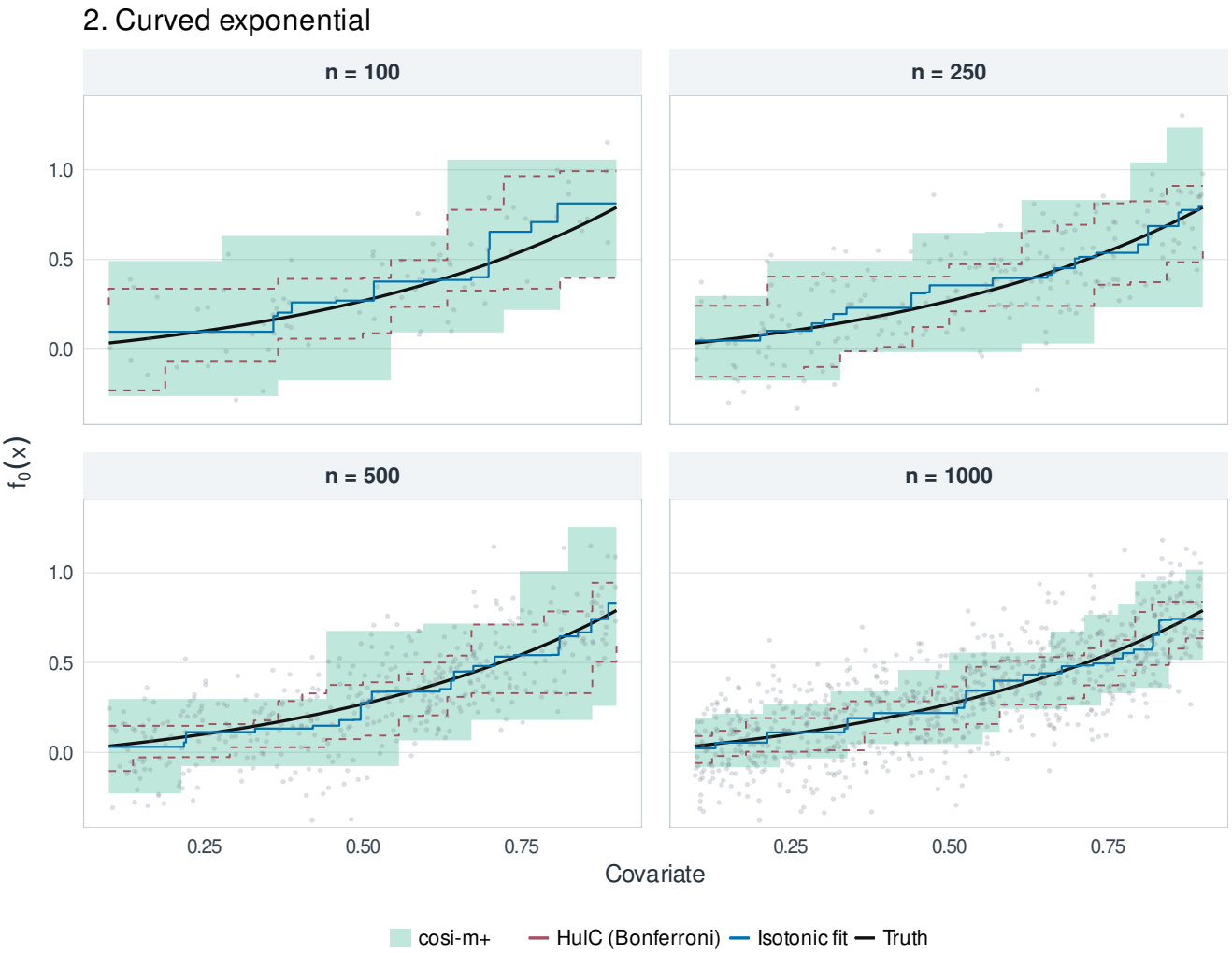


Figure 2: Setting 2: Exponential mean, uniform covariates and Gaussian errors with standard deviation 0.2.

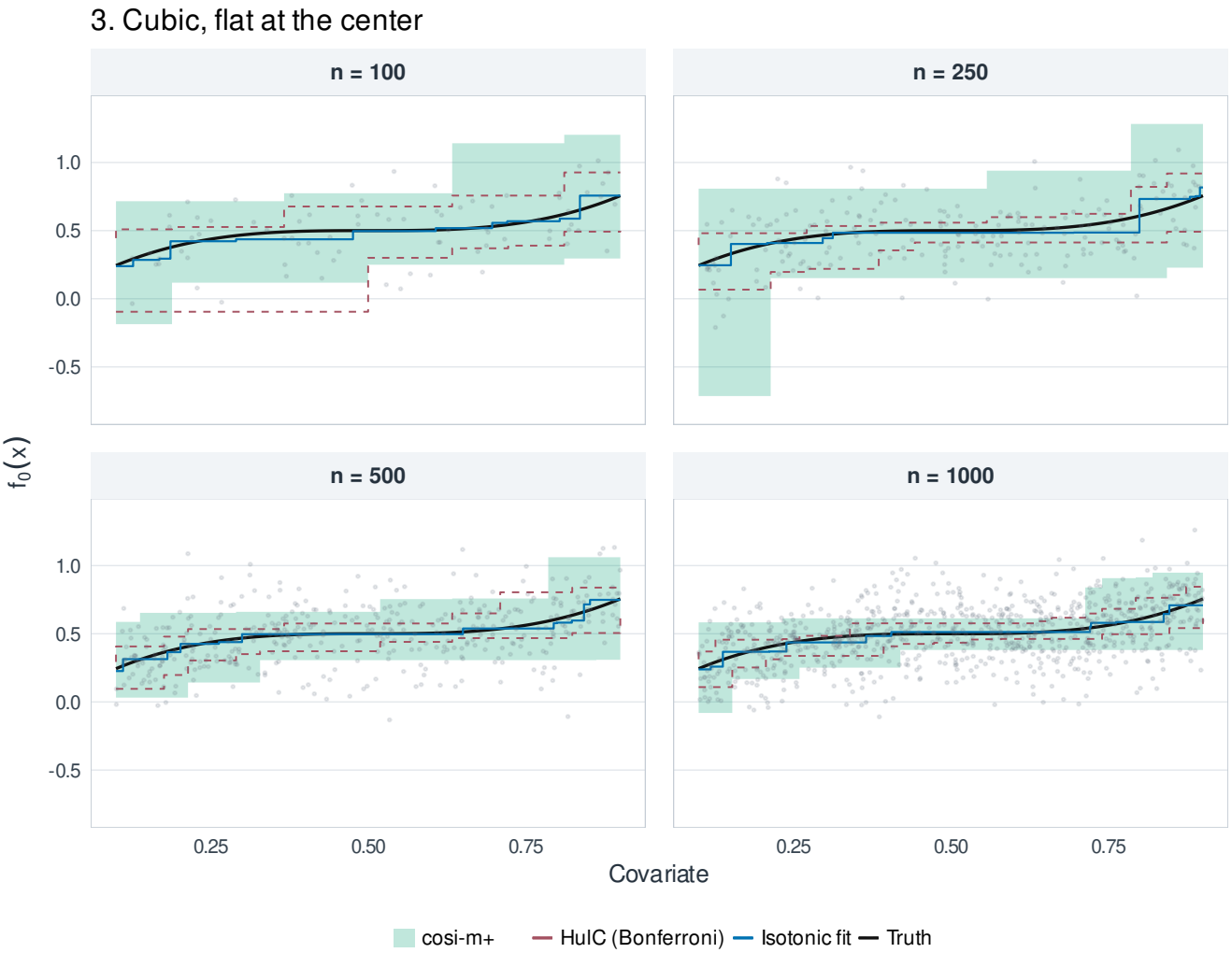


Figure 3: Setting 3: Cubic mean with zero derivative at 1/2, uniform covariates and Gaussian errors.

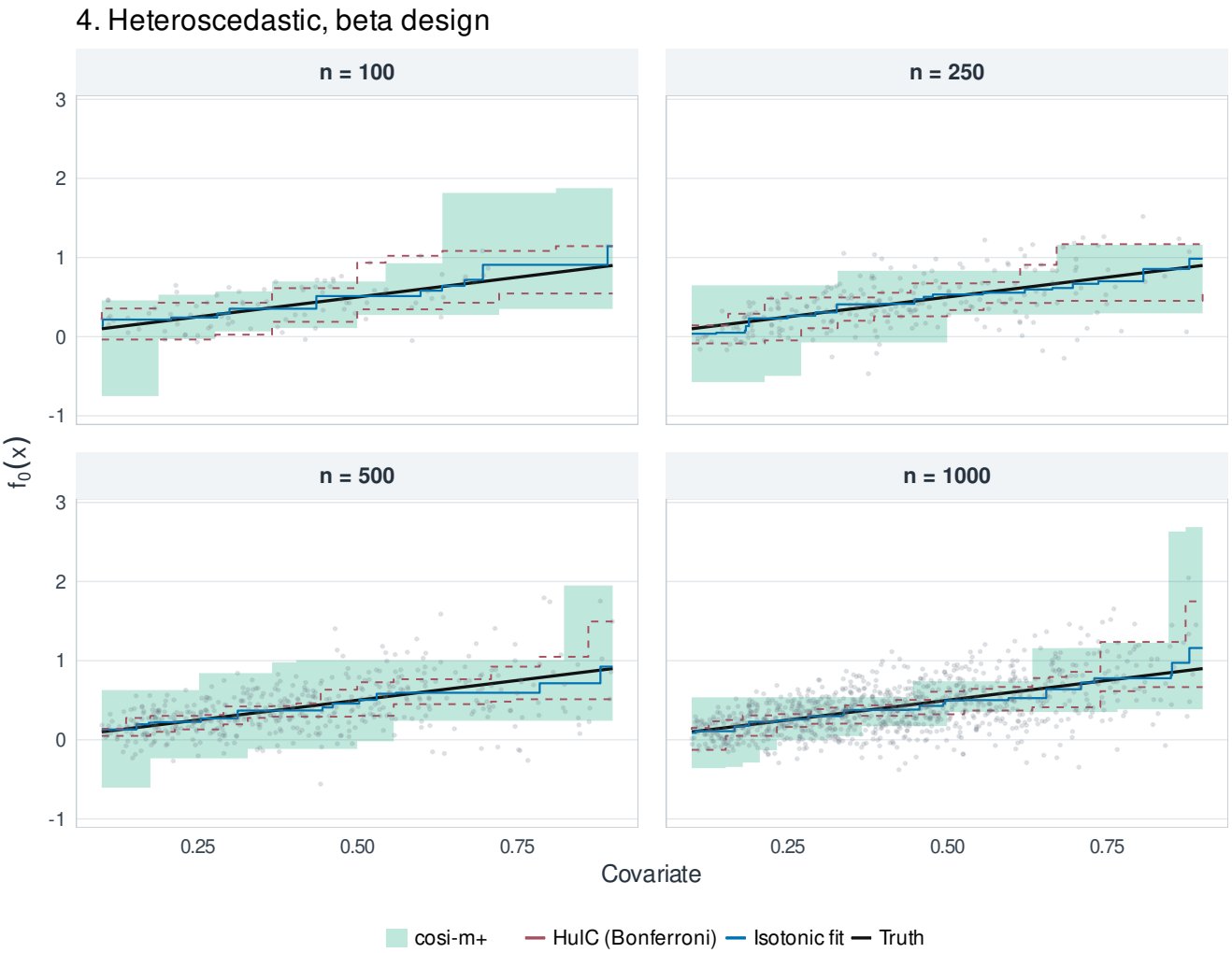


Figure 4: Setting 4: Linear mean with Beta$(2, 3)$ covariates and noise standard deviation $0.1 + 0.4x$.

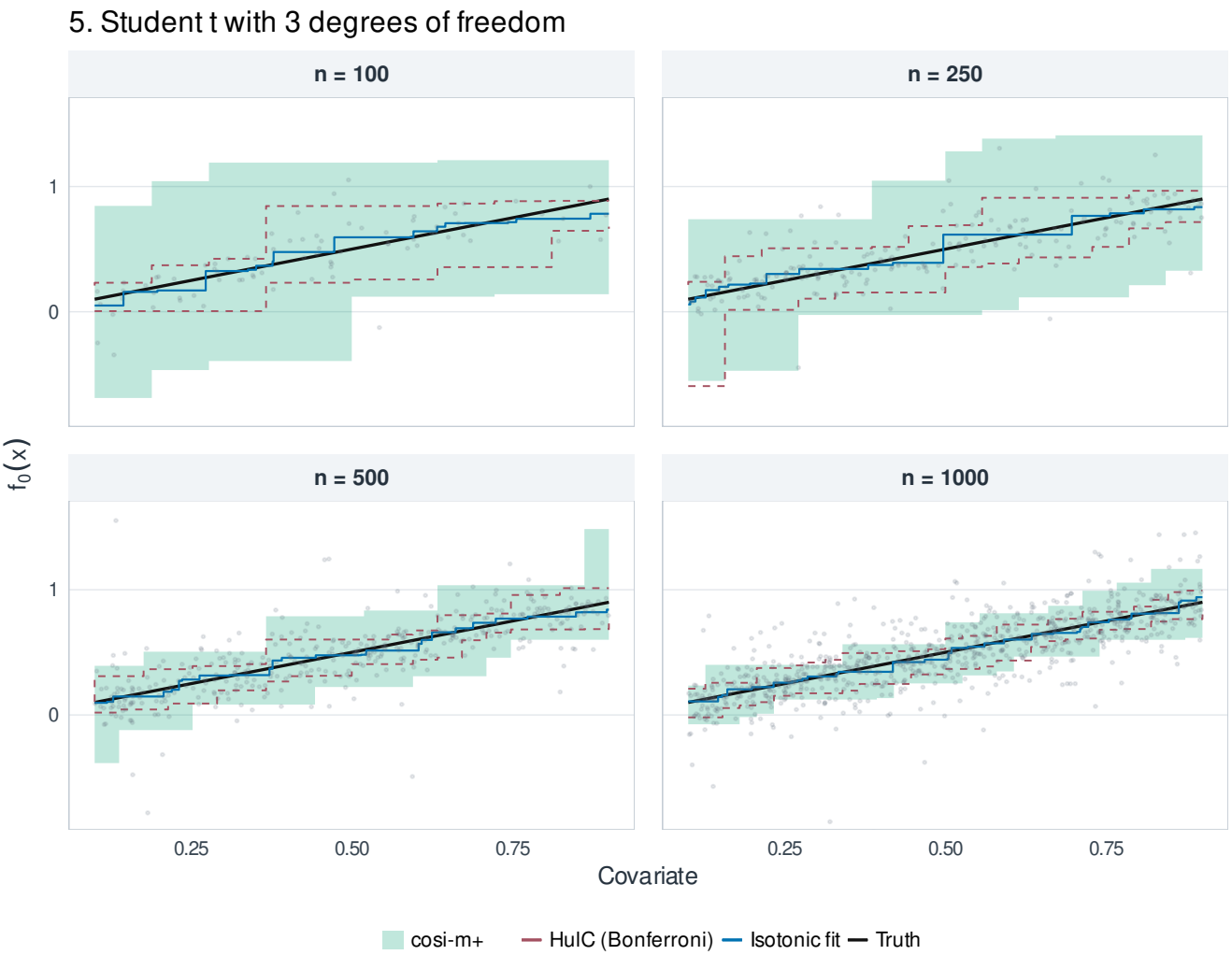


Figure 5: Setting 5: Linear mean, uniform covariates and variance-normalized Student t_3 errors.

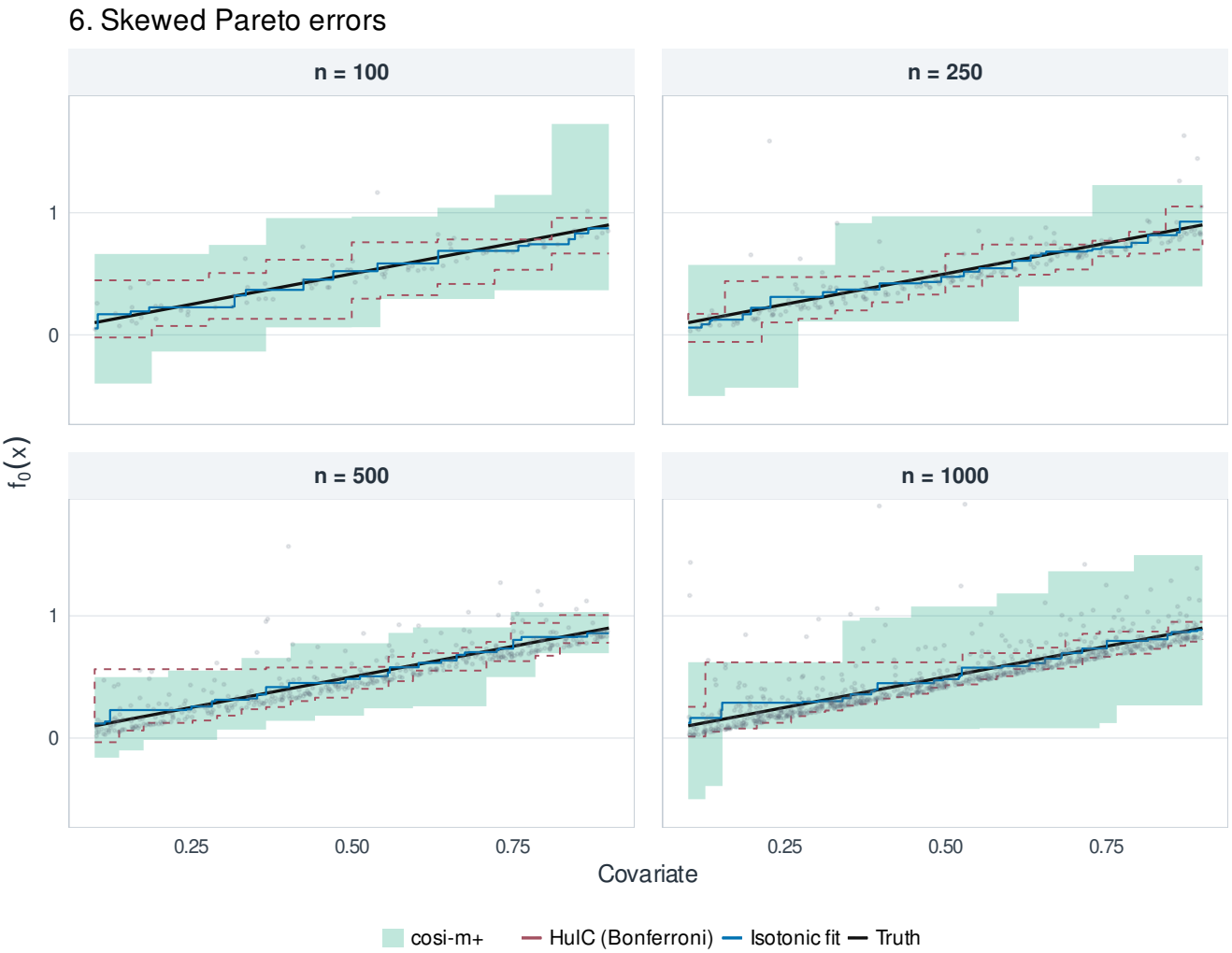


Figure 6: Setting 6: Linear mean, uniform covariates and centered, variance-normalized Pareto errors of shape 2.5.

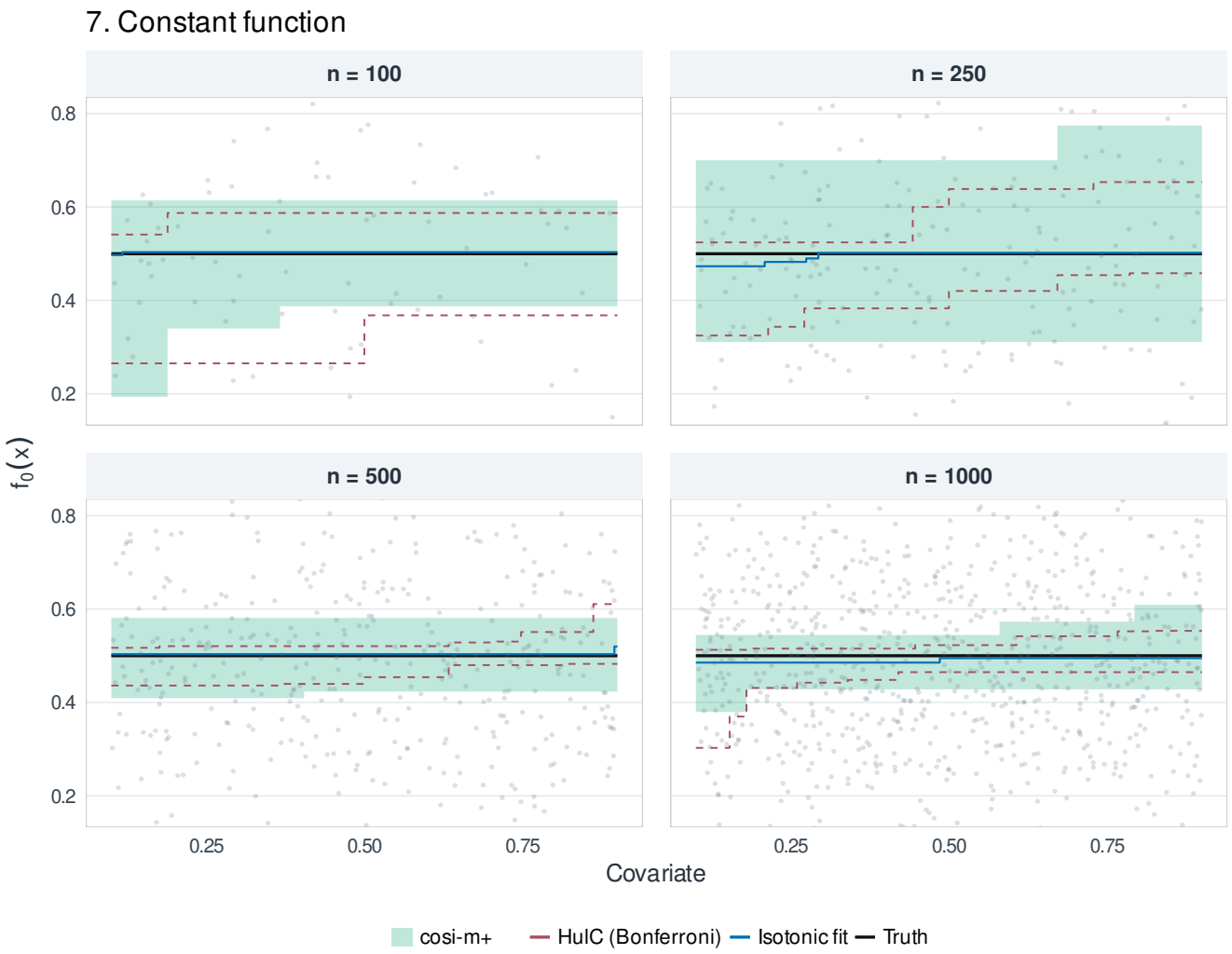


Figure 7: Setting 7: Constant mean $f_0 = 1/2$, uniform covariates and Gaussian errors.

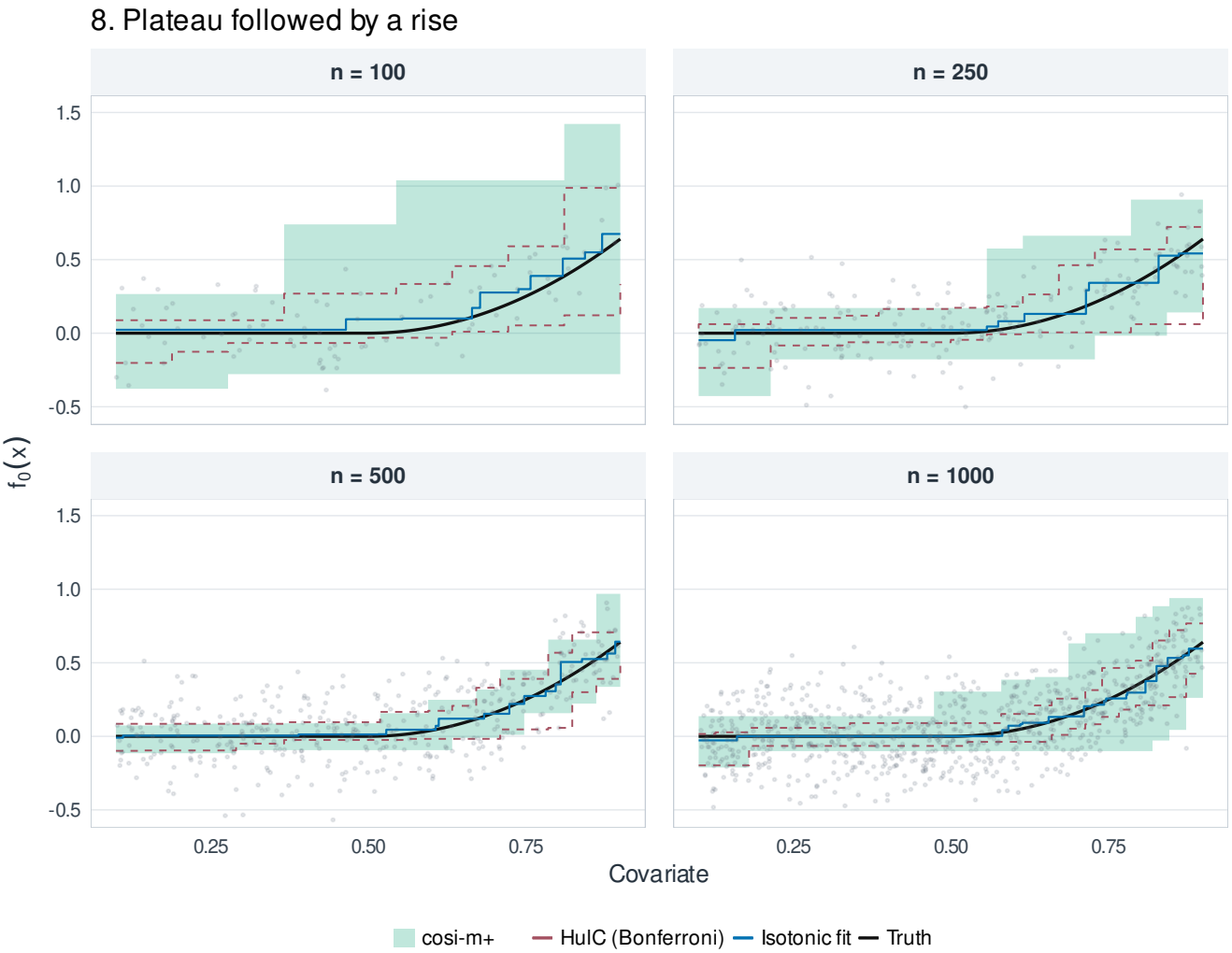


Figure 8: Setting 8: Flat region followed by a quadratic rise, with uniform covariates and Gaussian errors.

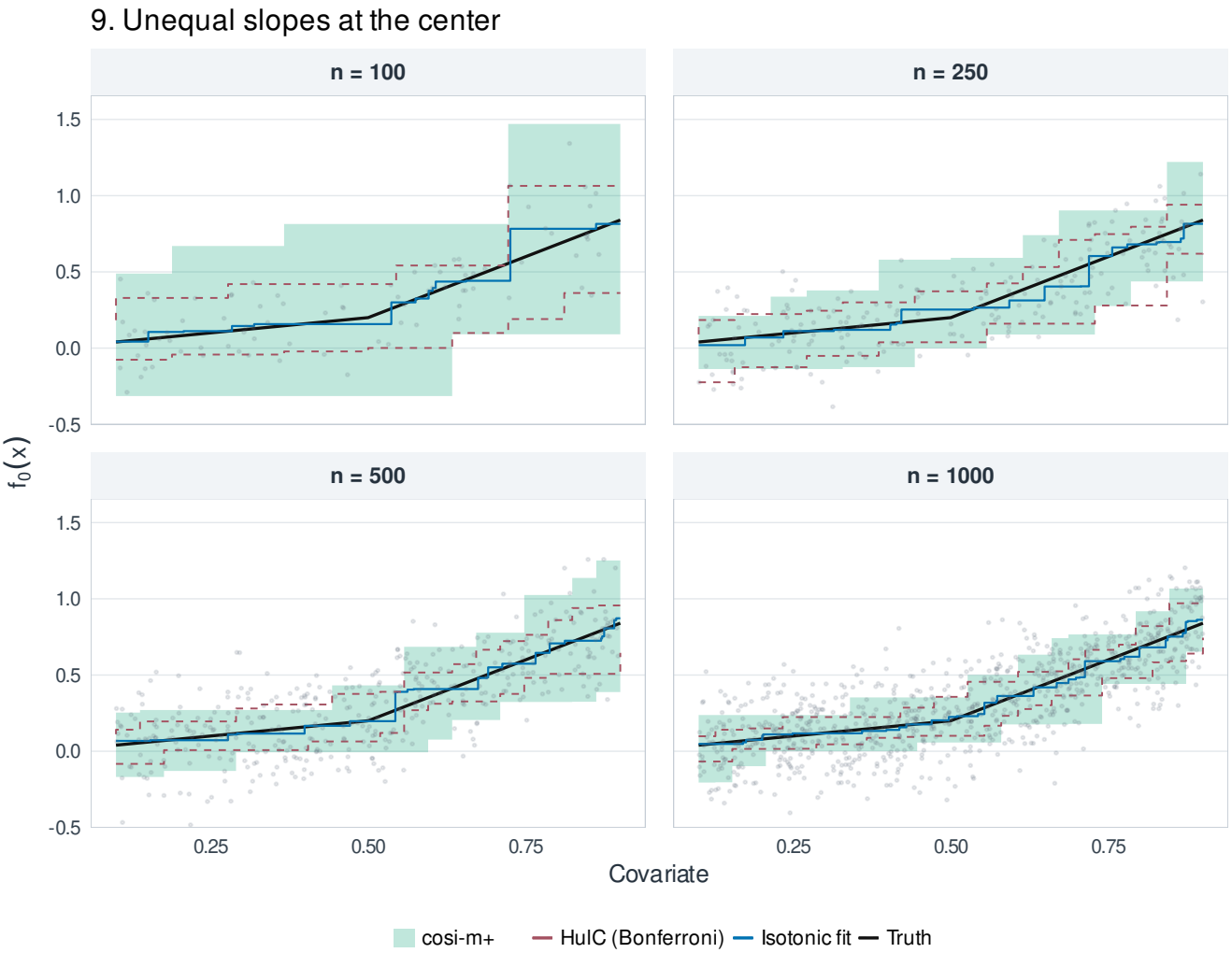


Figure 9: Setting 9: Continuous kink with left and right slopes 0.4 and 1.6, with uniform covariates and Gaussian errors.

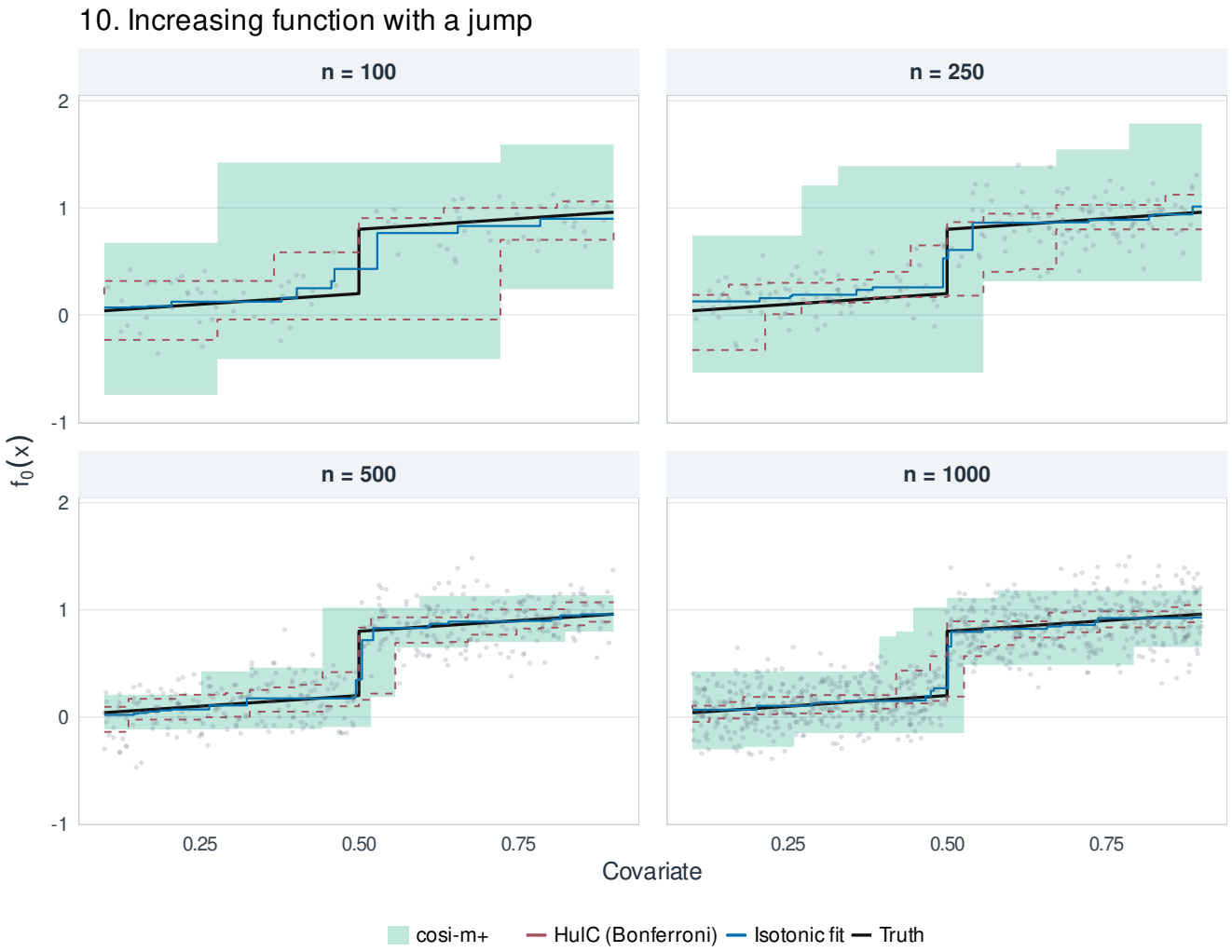


Figure 10: Setting 10: An increasing mean with a jump at $1/2$, where $f_0(1/2) = 0.8$, with uniform covariates and Gaussian errors.

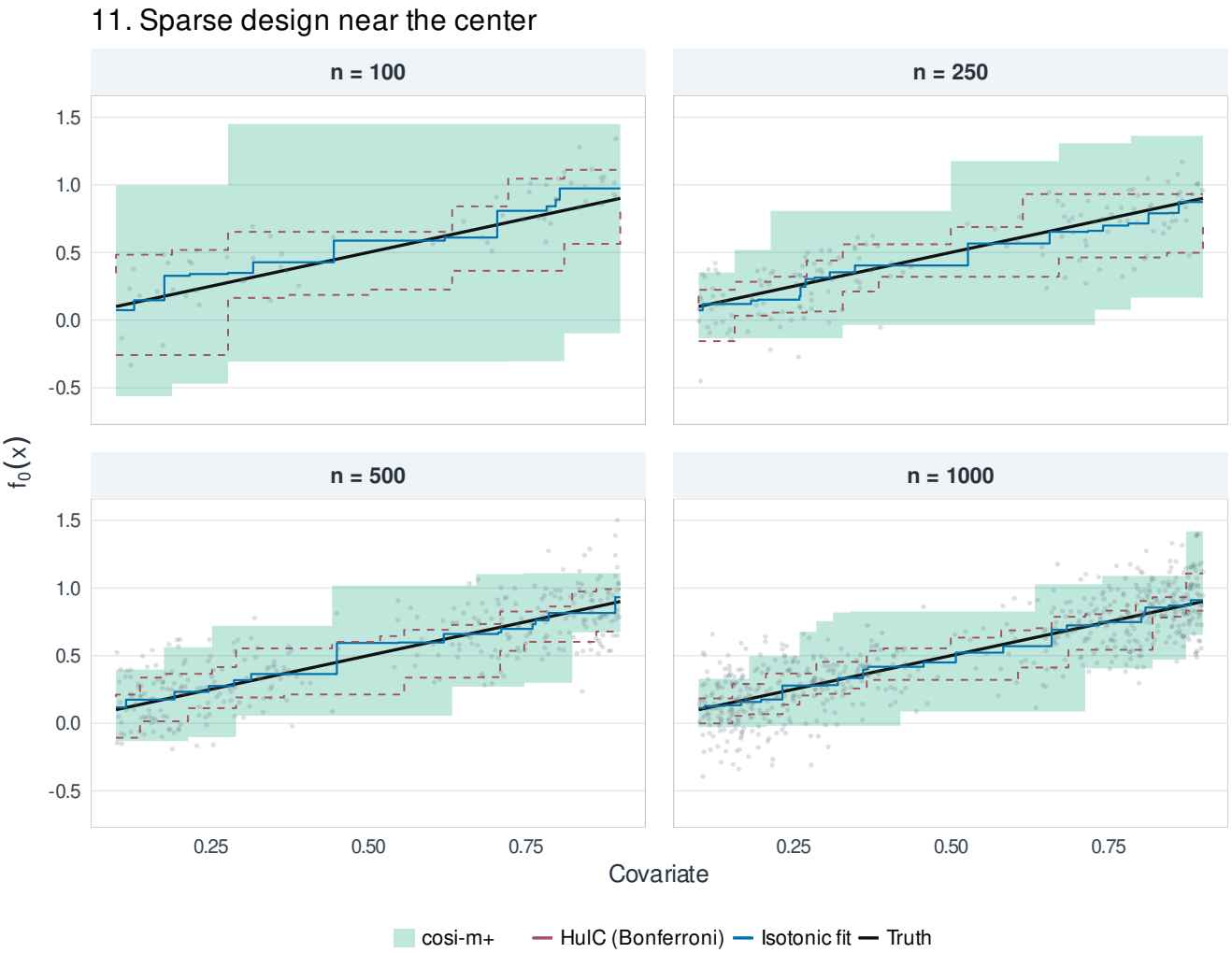


Figure 11: Setting 11: Linear mean with covariate density $4|x - 1/2|$, which vanishes at the center, and Gaussian errors.

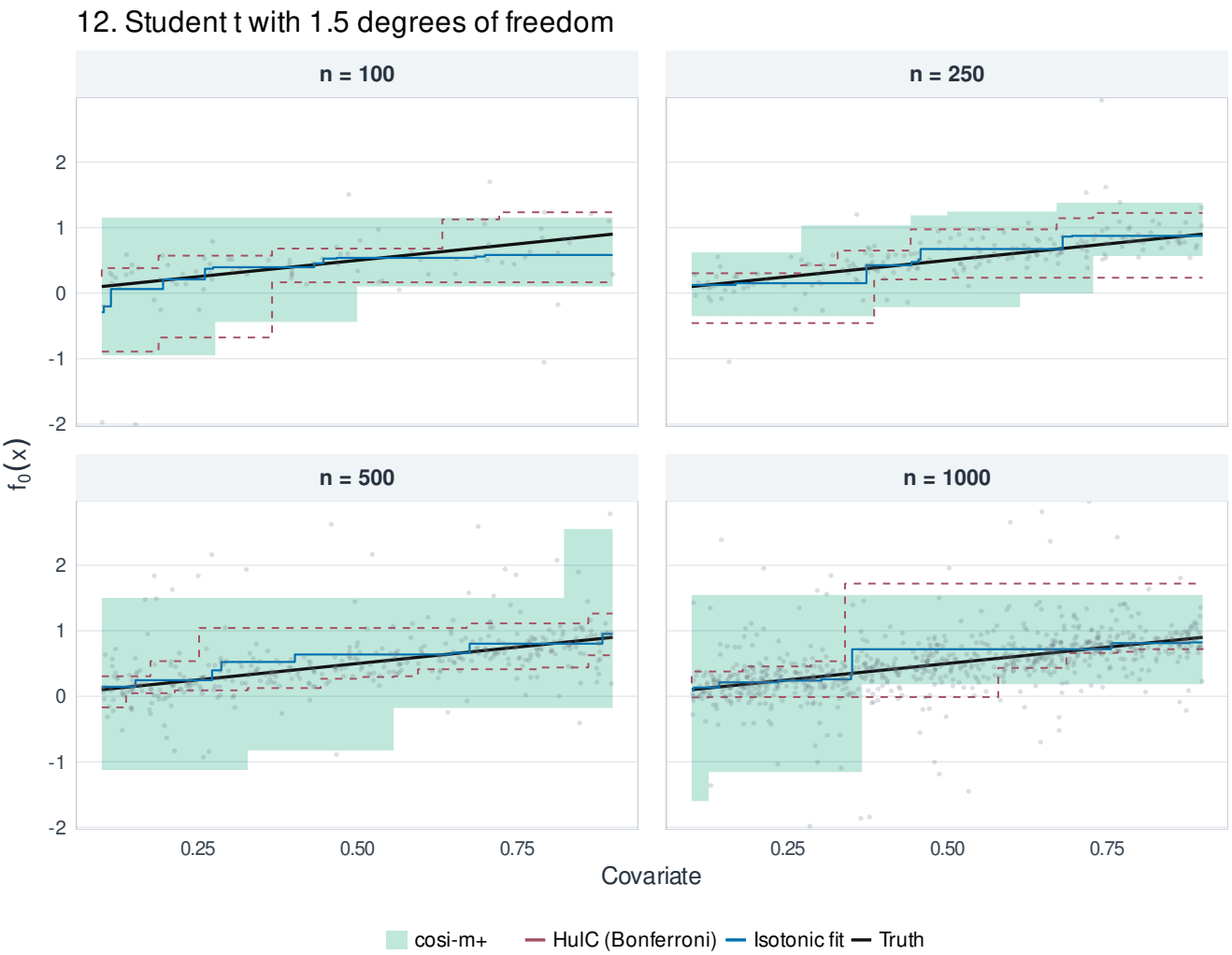


Figure 12: Setting 12: Linear mean, uniform covariates and Student $t_{1.5}$ errors scaled to match the Gaussian interquartile range.

2 Isotonic fits and the two COSI rectangles

For each sample size, take $K = \lfloor \sqrt{n} \rfloor$ anchors equally spaced on $[0.1, 0.9]$ and replace the anchor nearest $1/2$ by $1/2$. This gives $K = 10, 15, 22, 31$, respectively. The replacement ensures that the zero-derivative point, kink, and jump are evaluated directly. Every method uses this same deterministic grid.

An isotonic least squares fit minimizes $\sum_i \{Y_i - f(X_i)\}^2$ over nondecreasing f. The implementation sorts the pairs by X, aggregates tied covariates using their counts, and applies the pool-adjacent-violators calculation. Predictions are right-continuous steps: between adjacent observed covariates, use the fitted value at the covariate to the left. Beyond the observed covariate range, clamp to the first or last fitted value. This convention is used for every split and resample, including bootstrap samples with repeated pairs.

Randomly permute the observations and reserve $n_0 = \lfloor n/2 \rfloor$ observations in S_0 for the pilot. Split the remaining observations into $B = 9$ balanced disjoint groups $S_1, \ldots, S_B$; this is the value returned by the calibration function `smallest_B` at $\alpha = 0.05$ and $\delta = 0.30$. Define

$$\vartheta_k = \widehat{f}_{S_0}(t_k), \qquad W_{j,k} = \widehat{f}_{S_j}(t_k), \qquad s_k = \frac{1}{\sqrt{N_{0,k}}},$$

where $N_{0,k}$ counts the original pilot observations in the constant PAVA block selected at t_k. At a clamped anchor, use the selected boundary block's count. At an observed covariate, this is the univariate selected-block count; at an off-design anchor we use the count associated with the stated step-extension convention. This is motivated by the selected-block quantity in (Deng et al., 2021, Section 1.1, equations (1.4) and (1.6)). We use s_k as a *block-count normalization*, without multiplying by a residual standard deviation. The standard-error expression in their equation (1.7) also includes a noise-scale estimate.

For clarity, the randomized cutoff $c^\star_{B,\beta}$ is implemented as follows. With $Z_B \sim \mathrm{Bin}(B, 1/2)$, set

$$\begin{aligned} q &= \min\{r : \mathbb{P}(Z_B > r) \le \beta\}, \\ p &= \frac{\mathbb{P}(Z_B > q-1) - \beta}{\mathbb{P}(Z_B = q)}, \\ c^\star_{B,\beta} &= q - \mathbf{1}\{U > p\}, \end{aligned}$$

where $U \sim \mathrm{Unif}(0,1)$. Draw $c_m = c^\star_{B,\alpha/2}$ and $c_+ = c^\star_{B,\alpha}$ independently, using one realization of each cutoff for all anchors. Both cutoffs take values in $\{6, 7\}$ in this study.

The rectangular cosi-m+ construction. Let $D_{j,k} = |W_{j,k} - \vartheta_k|/s_k$, and let $D_{(r),k}$ denote the rth order statistic over the B splits at coordinate k. Set

$$R_+ = \max_{1 \le k \le K} D_{(c_+ + 1),k}, \qquad I^+_k = [\vartheta_k - R_+ s_k,\ \vartheta_k + R_+ s_k].$$

The order statistic is taken *before* the maximum over coordinates. In particular, this is not the order statistic of the splitwise maximum distances.

The rectangular enclosure of cosi-m. Set $L_k = W_{(B-c_m),k}$ and $U_k = W_{(c_m+1),k}$. The selected-coordinate set uses $k_\theta = \min \arg\max_k |\vartheta_k - \theta_k|/s_k$ and the condition $\theta_{k_\theta} \in [L_{k_\theta}, U_{k_\theta}]$. We first take its closed coordinatewise rectangular enclosure. Write

$$r_k = \frac{\max\{|L_k - \vartheta_k|, |U_k - \vartheta_k|\}}{s_k}, \qquad R_{-k} = \max_{\substack{\ell \ne k \\ r_\ell > 0 \text{ or } \ell = 1}} r_\ell,$$

with $\max \varnothing = -\infty$. The enclosure has endpoints

$$\ell^m_k = \min\{L_k, \vartheta_k - s_k R_{-k}\}, \qquad u^m_k = \max\{U_k, \vartheta_k + s_k R_{-k}\}.$$

The eligibility condition handles the degenerate zero-radius tie. In the figures, `cosi-m (rectangle)` denotes this enclosure. Its reported width is the width of the associated monotone band; the nonrectangular selected-coordinate set is not used to calculate these widths. Because c_m and c_+ use different levels and separate draws, the two reported rectangles need not be nested.

3 From anchor rectangles to bands and width measurements

For any method's anchor endpoints (ℓ_k, u_k), apply monotone tightening

$$A_k = \max_{i \le k} \ell_i, \qquad D_k = \min_{i \ge k} u_i.$$

At t_k, the band is $[A_k, D_k]$; for $t_k < x < t_{k+1}$, it is $[A_k, D_{k+1}]$. To the left of t_1 the lower endpoint is $-\infty$, and to the right of t_K the upper endpoint is $+\infty$. This uses the envelope construction of Kuchibhotla et al. (2024); the explicit two-anchor formula is equation (35), Section 5.3, pp. 27–28 of arXiv:2105.14577v2.

The finite width reported in the comparisons is the average height over the anchor span:

$$\overline{w} = \frac{1}{0.8} \sum_{k=1}^{K-1} (t_{k+1} - t_k)(D_{k+1} - A_k).$$

It is not the average of the K pointwise widths. The unbounded tails are outside the displayed and integrated domain $[0.1, 0.9]$. If any $A_k > D_k$, the band is flagged as empty, its recorded width is zero, and its coverage indicator is false. No replication is discarded or redrawn. The empirical simultaneous-coverage indicator checks all K tightened anchor intervals. Width ratios divide $\overline{w}$ by the HulC band's width from the same replication, before computing boxplots or medians.

4 Full-sample comparison procedures

The four benchmarks use all n observations. Let $\widehat{f}_n(t_k)$ be the full-sample isotonic estimate. The bootstrap and subsampling calculations use the same pilot scales s_k as COSI, held fixed during resampling. All supremum-root critical values use R's `quantile(type=6)`.

Pairs bootstrap. Draw 500 samples of n pairs with replacement and refit isotonic regression. For each bootstrap fit $\widehat{f}_n^{*r}$, compute

$$T_r^* = \max_k \frac{|\widehat{f}_n^{*r}(t_k) - \widehat{f}_n(t_k)|}{s_k}.$$

If q^* is the empirical 0.95 quantile, use the anchor intervals $[\widehat{f}_n(t_k) \pm q^* s_k]$, followed by the same monotone envelope. This is a supremum-distance pairs-bootstrap band, not coordinatewise percentile intervals.

Subsampling with rate 1/3. Draw 500 subsets of $b = \text{round}(n^{2/3})$ pairs without replacement. Each subset produces an isotonic fit $\widehat{f}_b^r$. If q_b is the empirical 0.95 quantile of $\max_k |\widehat{f}_b^r(t_k) - \widehat{f}_n(t_k)|/s_k$, use the intervals $[\widehat{f}_n(t_k) \pm (b/n)^{1/3} q_b s_k]$ and then tighten them. The resampling structure follows (Politis et al., 1999, Chapter 2, pp. 39–64).

Subsampling with estimated coordinatewise rates. Use 200 additional random subsets at each distinct feasible size $b_j = \text{round}(n^{\beta_j})$, with $\beta_j \in \{0.9, 0.75, 0.6, 0.45\}$ and $8 \le b_j < n$. At each anchor, calculate four quantile ranges using probability pairs $(0.1, 0.6)$, $(0.2, 0.7)$, $(0.3, 0.8)$, and $(0.4, 0.9)$. The quantiles are obtained by inverting the empirical CDF on a 100-point grid spanning the observed values. Regress the mean of the four log ranges on $\log b_j$; the negative slope is $\widehat{\gamma}_k$. This follows the quantile-range approach of Bertail et al. (1999); see also (Politis et al., 1999, Chapter 8, pp. 173–187). The numerical grids and resample counts are the choices made for this simulation.

Reuse the same 500 construction subsets as the fixed-rate method and form

$$\widehat{T}_r = \max_k (b/n)^{\widehat{\gamma}_k} \frac{|\widehat{f}_b^r(t_k) - \widehat{f}_n(t_k)|}{s_k}.$$

Thus the rate adjustment precedes the maximum. With $\widehat{q}$ its empirical 0.95 quantile, use $[\widehat{f}_n(t_k) \pm \widehat{q} s_k]$ and tighten. Finite estimated rates, including negative values, are retained without clipping. A degenerate quantile range, a nonfinite regression result, or fewer than two usable sizes triggers a flagged fallback $\widehat{\gamma}_k = 0$ at that coordinate.

HulC with a Bonferroni allocation. Use one balanced random partition of the full sample into $B_H = \lceil \log_2(2K/\alpha) \rceil$ groups, giving $B_H = 9, 10, 10, 11$ at the four sample sizes. At each anchor, take the minimum and maximum of the B_H isotonic estimates and apply the monotone envelope. The number of groups is deterministic; it is not randomized between adjacent integers.

Reproducibility and saved records. For setting s, sample size n, and replication r, the data seed is $100 + 10^6 s + 10^3 n + r$. The COSI split/cutoff stream uses this seed plus 5×10^7, and the benchmark stream uses it plus 10^8. R's random-number algorithms are fixed to `Mersenne-Twister`, `Inversion`, and `Rejection`. Every method uses the same data within a replication. The illustrations use the prespecified first replication, $r = 1$, at each sample size. The companion materials include band records, rate diagnostics, seeds, and source hashes. At 95% coverage, 500 independent replications give a Monte Carlo standard error of approximately one percentage point. Coverage figures use panel-specific lower limits and a common upper limit of 101%, retaining every empirical coverage value. The vertical scales are labeled separately in each panel.

5 Observed coverage and width

The study comprises 24,000 simulated datasets and 144,000 method-level band records. Each setting and sample-size combination includes 500 replications for every method. No empty bands occurred, and no datasets were redrawn or removed.

Across the 44 cells with a continuous mean function, empirical coverage of cosi-m+ ranges from 97.2% to 100.0%; at $n = 1000$, it ranges from 98.4% to 100.0%. The corresponding range for the cosi-m rectangular enclosure over all four sample sizes is 98.2% to 100.0%. These values describe simultaneous containment of all anchors and their monotone envelope. The two COSI bands are frequently conservative in the tested continuous settings. Their widths must therefore be read together with their coverage.

For the linear Gaussian setting, the median average width of cosi-m+ decreases from 1.032 at $n = 100$ to 0.440 at $n = 1000$. Its median paired width ratio relative to HulC changes from 1.962 to 1.917. The COSI methods reserve half of the sample for the pilot; the HulC benchmark fits its blocks using the full sample. The comparison thus includes the cost of that pilot allocation. Across the continuous settings, the median paired cosi-m+/HulC ratios range from 1.551 to 2.336. Table 3 reports the largest-sample widths and paired ratios.

The benchmark methods have distinct empirical behavior. In the linear Gaussian setting at $n = 1000$, coverage is 95.2% for pairs bootstrap, 96.8% for fixed-rate subsampling, 80.6% for estimated-rate subsampling, and 95.8% for HulC. Under skewed Pareto errors, the corresponding values are 97.8%, 92.2%, 99.2%, and 83.8%, compared with 99.8% for cosi-m+. These comparisons apply to the resampling and splitting implementations specified above.

Flatness and local asymmetry are visible in the sign diagnostics. For the constant function at $n = 1000$, the fraction of inference-block estimates above the truth ranges from 0.195 to 0.780 across the anchors; at $1/2$ it is 0.486. For the unequal-slope setting, the center fraction is 0.608. The full sign-frequency curves are provided in the companion diagnostic figures. These finite-sample sign frequencies are measured directly, without imposing median balance on the split estimates.

The jump setting behaves differently. cosi-m+ coverage changes from 88.2% at $n = 100$ to 70.2% at $n = 1000$; HulC coverage is 0.0% at the largest sample size. The target at the jump is $f_0(1/2) = 0.8$, while the off-design prediction convention selects the fitted value at the last observed covariate at or below $1/2$. This is a deliberate discontinuity and point-value stress experiment. Its numerical results are sensitive to that convention and should be interpreted as such.

The estimated-rate comparison retains a separate exponent at every anchor. An explicit zero-rate fallback was used at one or more anchors in 2.1% of replications. The companion diagnostics include coordinatewise median rates, their 10th and 90th percentiles, and the fallback frequency. No finite exponent was truncated to a prescribed range.

5.1 Interpreting the pairs-bootstrap results

Ordinary pairs resampling of the isotonic least squares estimator is inconsistent in the usual pointwise cube-root setting; see (Groeneboom and Jongbloed, 2024, Section 1). The linear Gaussian setting already belongs to this regime, so unusual error distributions are not needed for this issue to arise. The empirical resampling scheme does not reproduce the deterministic local drift governing the sampling distribution. Good coverage in some finite-sample experiments does not resolve this distributional problem.

The present comparison assesses a symmetric maximum-absolute-distance band over $K = \lfloor \sqrt{n} \rfloor$ anchors at a single nominal level. It does not compare conditional pointwise bootstrap distributions with their sampling distributions. Failure to estimate a distribution consistently does not require substantial undercoverage at every confidence level and sample size. Taking absolute values and a maximum changes the statistic being assessed; those operations do not establish consistency of the bootstrap band. Across the continuous-function settings, its empirical coverage ranges from 90.8% (setting 4, $n = 100$) to 99.2% (setting 7, $n = 1000$). In setting 11 it ranges from 92.4% to 93.2% across the four sample sizes. Thus its performance includes undercoverage as well as conservatism.

The jump setting gives bootstrap coverage between 24.8% and 32.6%, but estimation at the discontinuity and the prediction convention are additional complications there. That example does not isolate bootstrap inconsistency. For monotone truth, tightening the anchor intervals preserves the simultaneous-containment event, so the monotone envelope itself cannot explain the favorable bootstrap coverage. A more direct distributional diagnostic would fix an interior point in the linear Gaussian setting and compare bootstrap and sampling quantiles over increasing sample sizes and several confidence levels. The current results describe the implemented band and do not establish that ordinary pairs bootstrap generally works for isotonic regression.

5.2 Interpreting the width comparison with HulC

HulC has the smaller median width in all 48 setting/sample-size cells, but there is no ordering in every replication. The cosi-m rectangular band is narrower than HulC in 278 of the 24,000 paired datasets (1.16%), and cosi-m+ is narrower in 387 (1.61%).

The comparison includes a substantial difference in data allocation. COSI reserves half the observations for the pilot, leaving approximately $n/18$ observations per inference block, whereas HulC uses approximately n/B_H per block, with $B_H \in \{9, 10, 11\}$. As a scale heuristic in a regular positive-slope cube-root regime, this suggests a factor $(18/B_H)^{1/3}$, between 1.18 and 1.26, for the pointwise estimation scales. This calculation is not a decomposition of the observed band widths and does not explain the full difference, which is often close to a factor of two.

The Bonferroni allocation is implemented through the number of HulC blocks, $B_H = \lceil \log_2(2K/\alpha) \rceil$, which changes only from 9 to 11 over the grids considered here. It does not multiply the band width by K. The Bonferroni label therefore does not determine the width ordering relative to these COSI constructions.

The constructions also differ geometrically. In cosi-m+, the maximum of the coordinatewise distance order statistics determines a common radius that is then applied at every anchor. HulC uses each anchor's own minimum and maximum. For cosi-m, the plotted band is based on the rectangular enclosure of the selected-coordinate set: the enclosure discards the set's restrictions coupling coordinates, and a large standardized radius at one anchor can enlarge intervals at other anchors. These features can contribute to wider bands; they do not imply deterministic containment of one method's band in another. All methods use the same monotone extension and the same integrated-width definition.

Finally, the empirical coverages are not matched. At $n = 1000$ in the linear Gaussian setting, HulC has median width 0.231 and coverage 95.8%, compared with median widths 0.464 and 0.440 for cosi-m and cosi-m+, both with 100% coverage. Under skewed Pareto errors, HulC and cosi-m+ have coverage 83.8% and 99.8%, respectively; for the constant function the corresponding values are 86.0% and 99.6%. The narrower HulC bands therefore reflect a combination of data allocation, band geometry, and different coverage behavior. The present simulation does not separate their individual contributions.

6 Numerical summaries at $n = 1000$

Setting	cosi-m	cosi-m+	HulC	Pairs	Sub. (1/3)	Sub. est.
1	100.0	100.0	95.8	95.2	96.8	80.6
2	100.0	100.0	96.4	95.4	95.6	79.2
3	100.0	100.0	92.6	96.4	98.6	85.2
4	100.0	99.8	94.8	94.6	97.4	95.6
5	99.4	99.6	97.6	95.2	95.6	92.6
6	100.0	99.8	83.8	97.8	92.2	99.2
7	100.0	99.6	86.0	99.2	100.0	96.6
8	100.0	100.0	90.8	93.8	98.4	87.8
9	100.0	100.0	96.8	93.6	96.6	82.2
10	72.4	70.2	0.0	24.8	52.8	19.4
11	100.0	99.8	91.4	92.4	93.6	82.6
12	98.6	98.4	96.6	96.2	93.4	99.4

Table 2: Empirical simultaneous coverage at $n = 1000$, in percent, from 500 replications per setting. cosi-m denotes its rectangular enclosure. The nominal level is 95%; the entries are empirical frequencies.

	Median average band width			Median paired ratio to HulC	
Setting	cosi-m+	HulC	Pairs	cosi-m	cosi-m+
1	0.440	0.231	0.242	2.001	1.917
2	0.419	0.221	0.232	2.013	1.914
3	0.352	0.183	0.188	2.063	1.942
4	0.675	0.343	0.398	2.078	1.961
5	0.432	0.224	0.245	1.988	1.921
6	0.402	0.192	0.239	2.144	2.081
7	0.161	0.093	0.060	1.928	1.753
8	0.359	0.186	0.183	2.054	1.951
9	0.435	0.225	0.233	2.024	1.930
10	0.537	0.218	0.215	2.735	2.474
11	0.537	0.267	0.286	2.145	2.030
12	0.967	0.573	0.580	1.712	1.656

Table 3: Widths at $n = 1000$. Average width is the integral of band height over $[0.1, 0.9]$, divided by 0.8. The last two columns are medians of within-replication ratios; they are not ratios of median widths. All methods' absolute widths and paired ratios are provided in the companion numerical summaries and figures.

7 Empirical coverage comparisons

Figures 13–15 compare simultaneous coverage as the sample size increases. Each panel retains all empirical values, with a separate vertical range and a common nominal reference of 95%. Coverage should be read together with the width comparisons that follow.

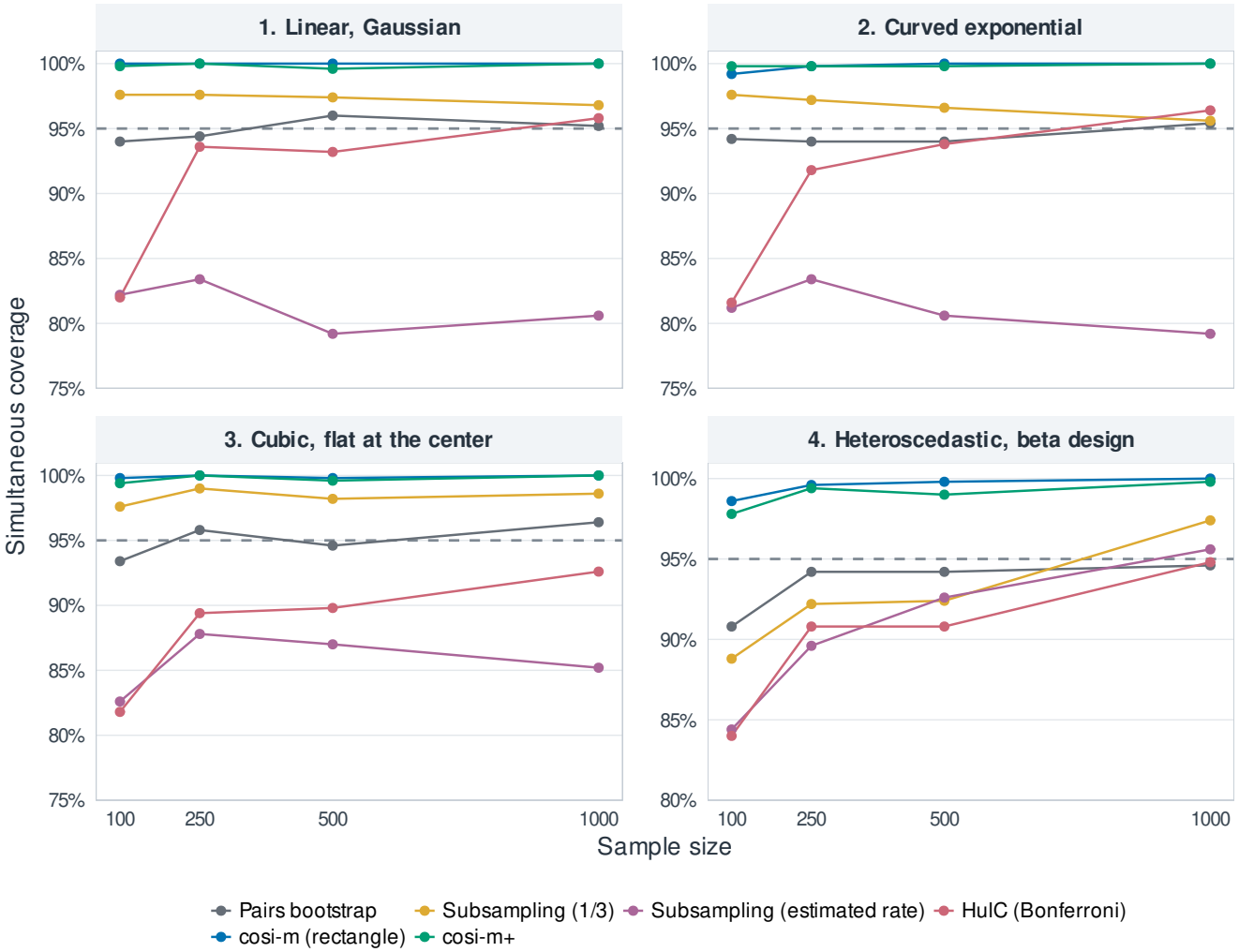


Figure 13: Empirical simultaneous coverage for settings 1–4. The same data are used for all methods within each of 500 replications at every sample size. The dashed line marks the nominal 95% level. Panel-specific vertical limits retain all coverage values. Each rectangle is extended using the same monotone envelope; every anchor is checked when recording coverage.

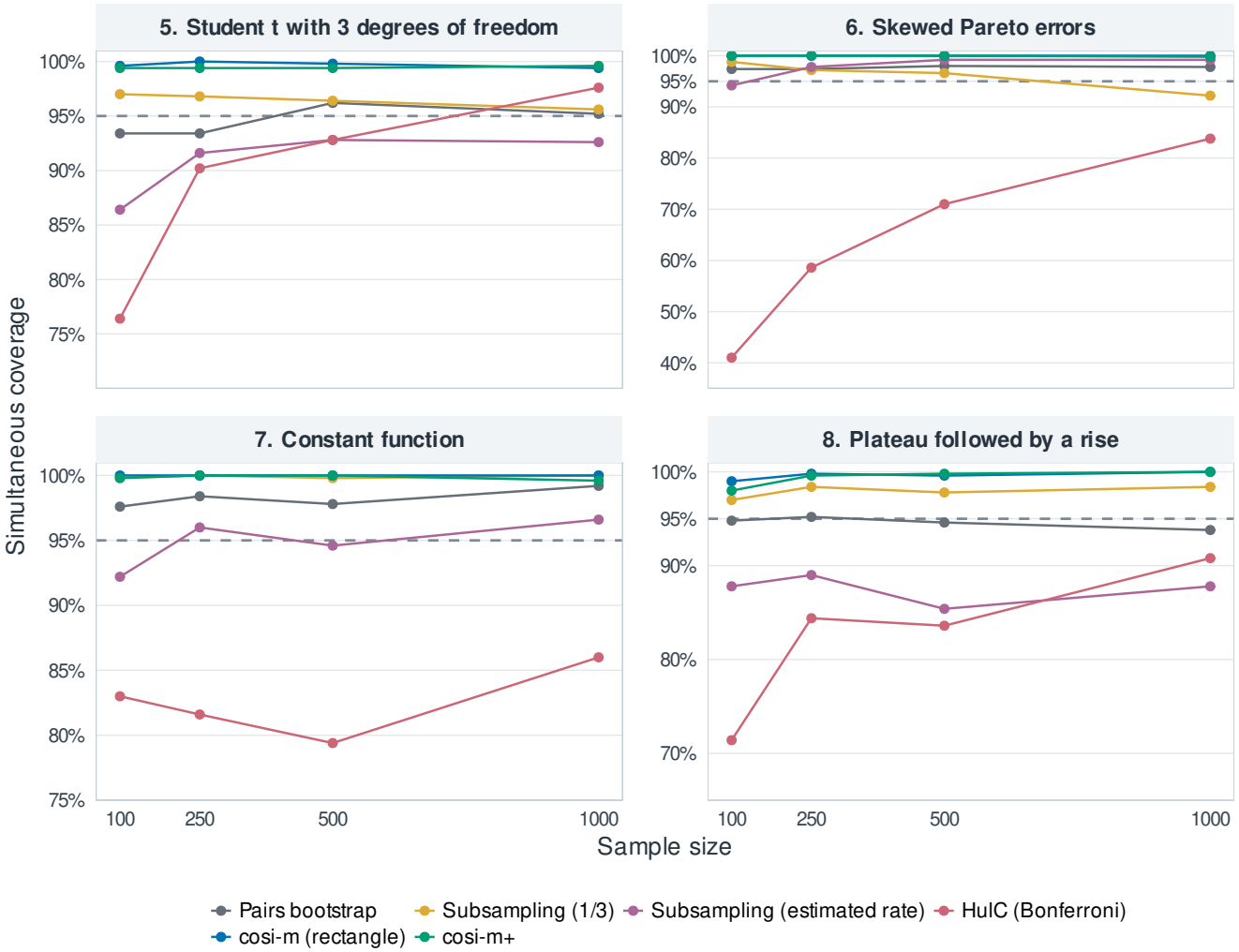


Figure 14: Empirical simultaneous coverage for settings 5–8. The same data are used for all methods within each of 500 replications at every sample size. The dashed line marks the nominal 95% level. Panel-specific vertical limits retain all coverage values. Each rectangle is extended using the same monotone envelope; every anchor is checked when recording coverage.

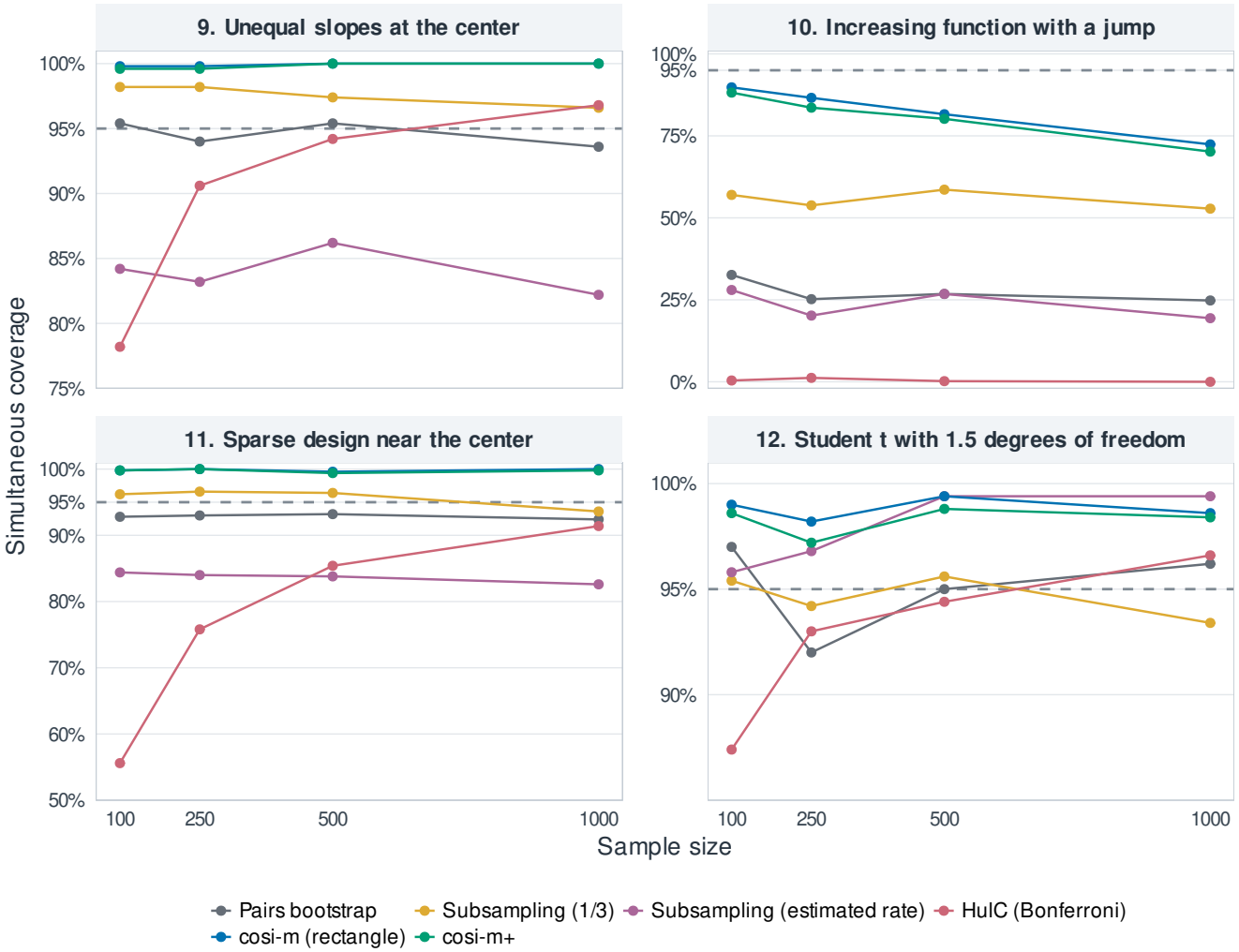


Figure 15: Empirical simultaneous coverage for settings 9–12. The same data are used for all methods within each of 500 replications at every sample size. The dashed line marks the nominal 95% level. Panel-specific vertical limits retain all coverage values. Each rectangle is extended using the same monotone envelope; every anchor is checked when recording coverage.

8 Band-width comparisons

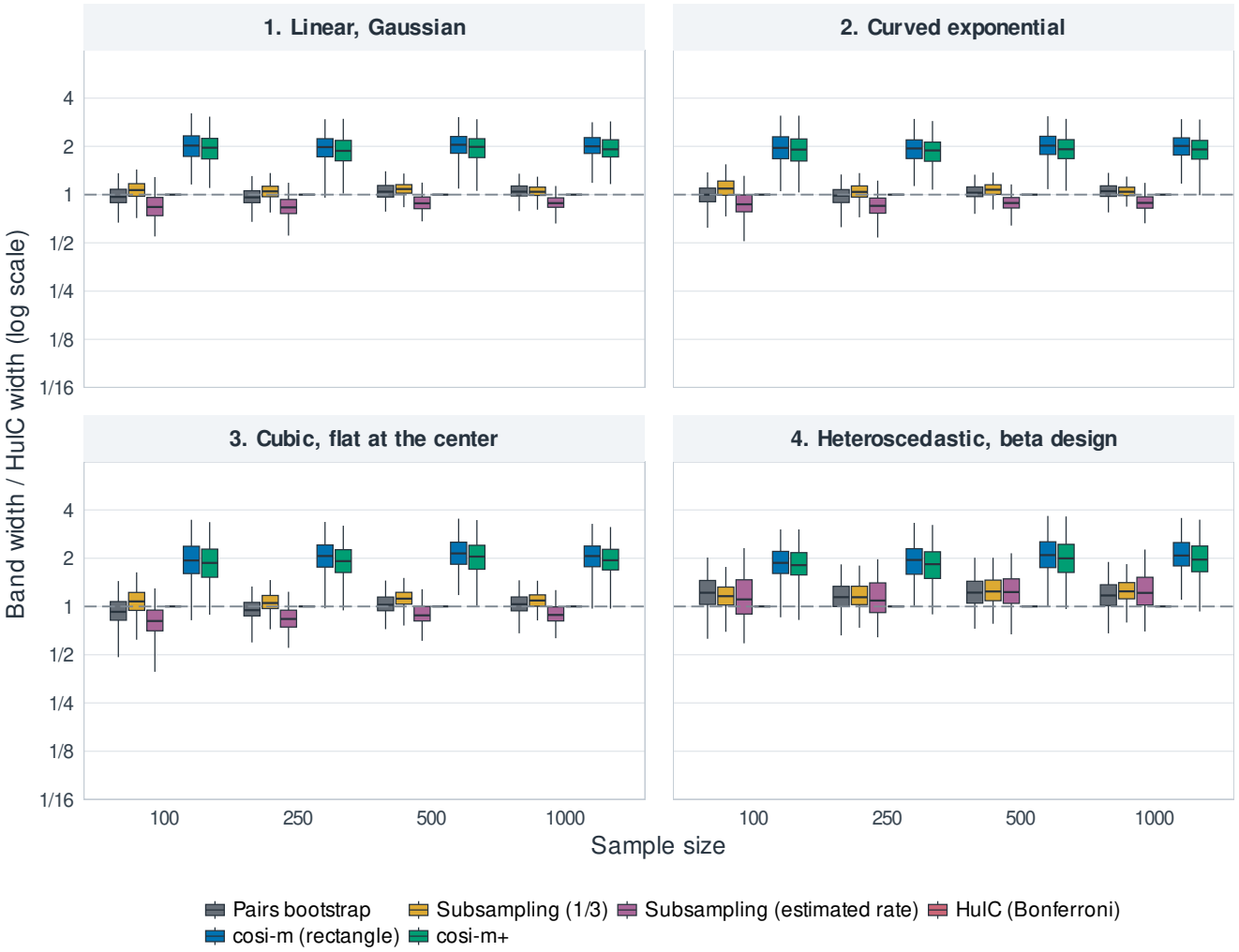


Figure 16: Paired ratios of average band width to the HulC width from the same data, for settings 1–4. Both average widths are measured on $[0.1, 0.9]$. The vertical axis is logarithmic and displays $[1/16, 8]$; the dashed line is one. Boxplot statistics are computed before restricting the displayed range. Outlier points are omitted. Companion figures display boxplots of absolute band width.

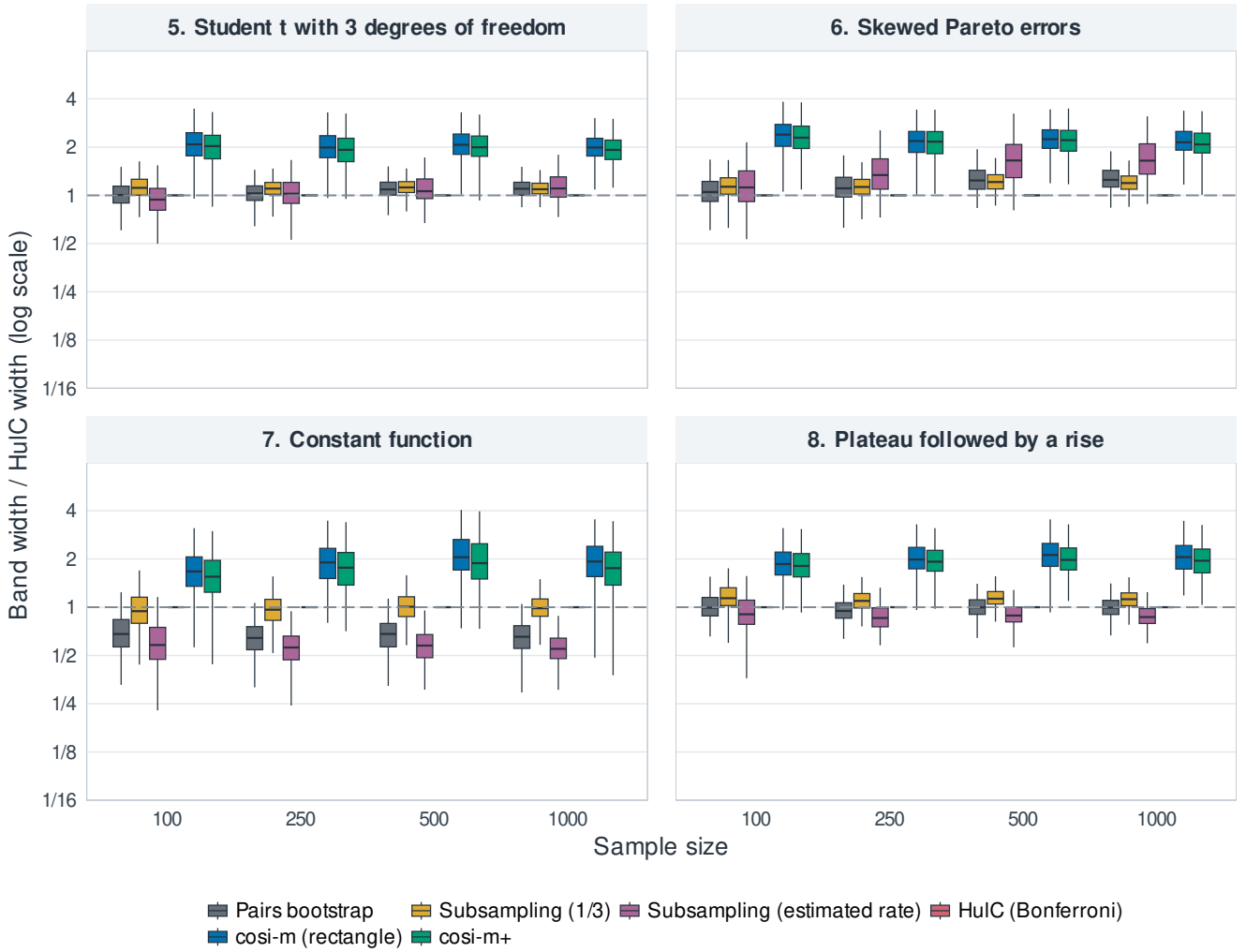


Figure 17: Paired ratios of average band width to the HulC width from the same data, for settings 5–8. Both average widths are measured on $[0.1, 0.9]$. The vertical axis is logarithmic and displays $[1/16, 8]$; the dashed line is one. Boxplot statistics are computed before restricting the displayed range. Outlier points are omitted. Companion figures display boxplots of absolute band width.

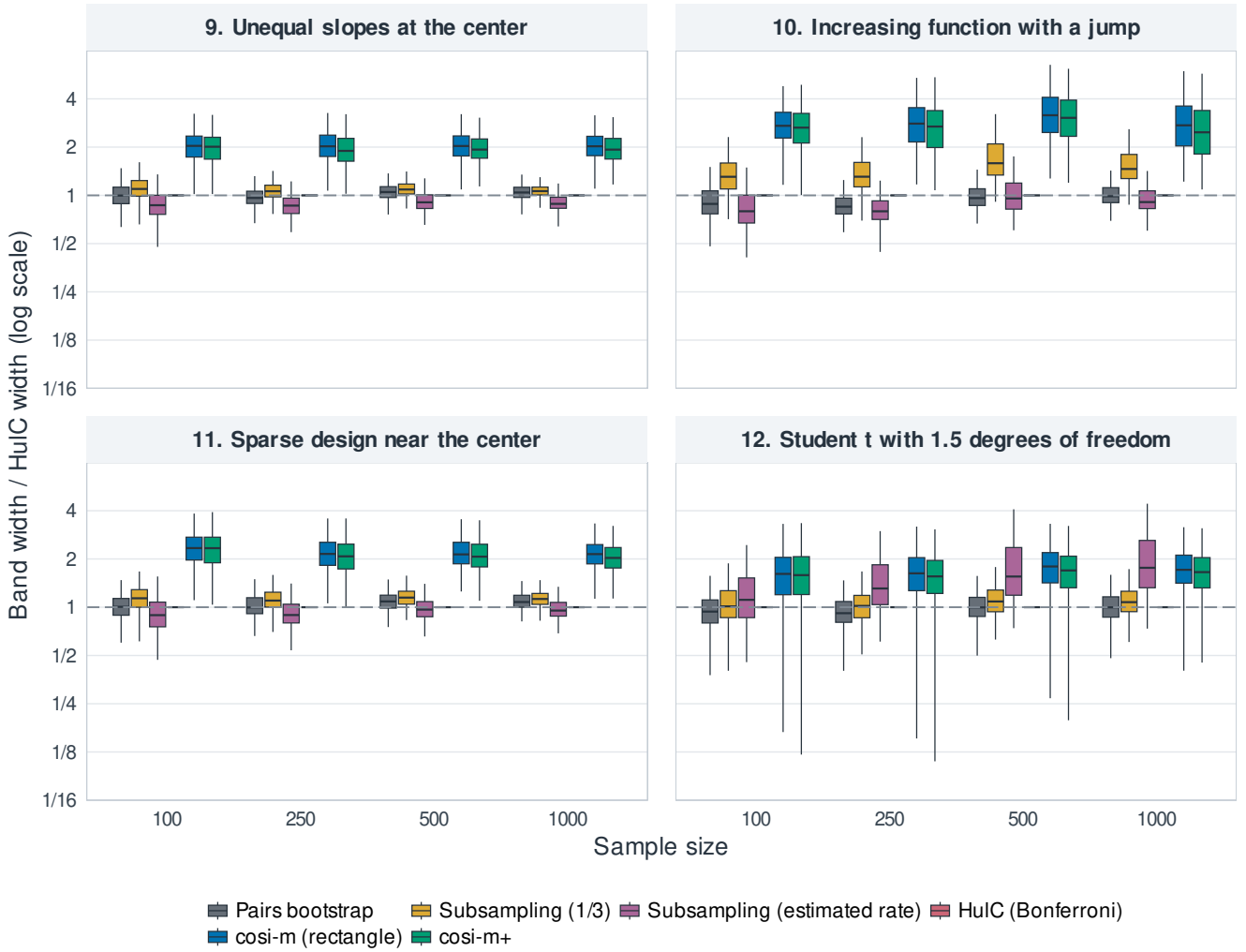


Figure 18: Paired ratios of average band width to the HulC width from the same data, for settings 9–12. Both average widths are measured on $[0.1, 0.9]$. The vertical axis is logarithmic and displays $[1/16, 8]$; the dashed line is one. Boxplot statistics are computed before restricting the displayed range. Outlier points are omitted. Companion figures display boxplots of absolute band width.

9 Empirical sign diagnostics

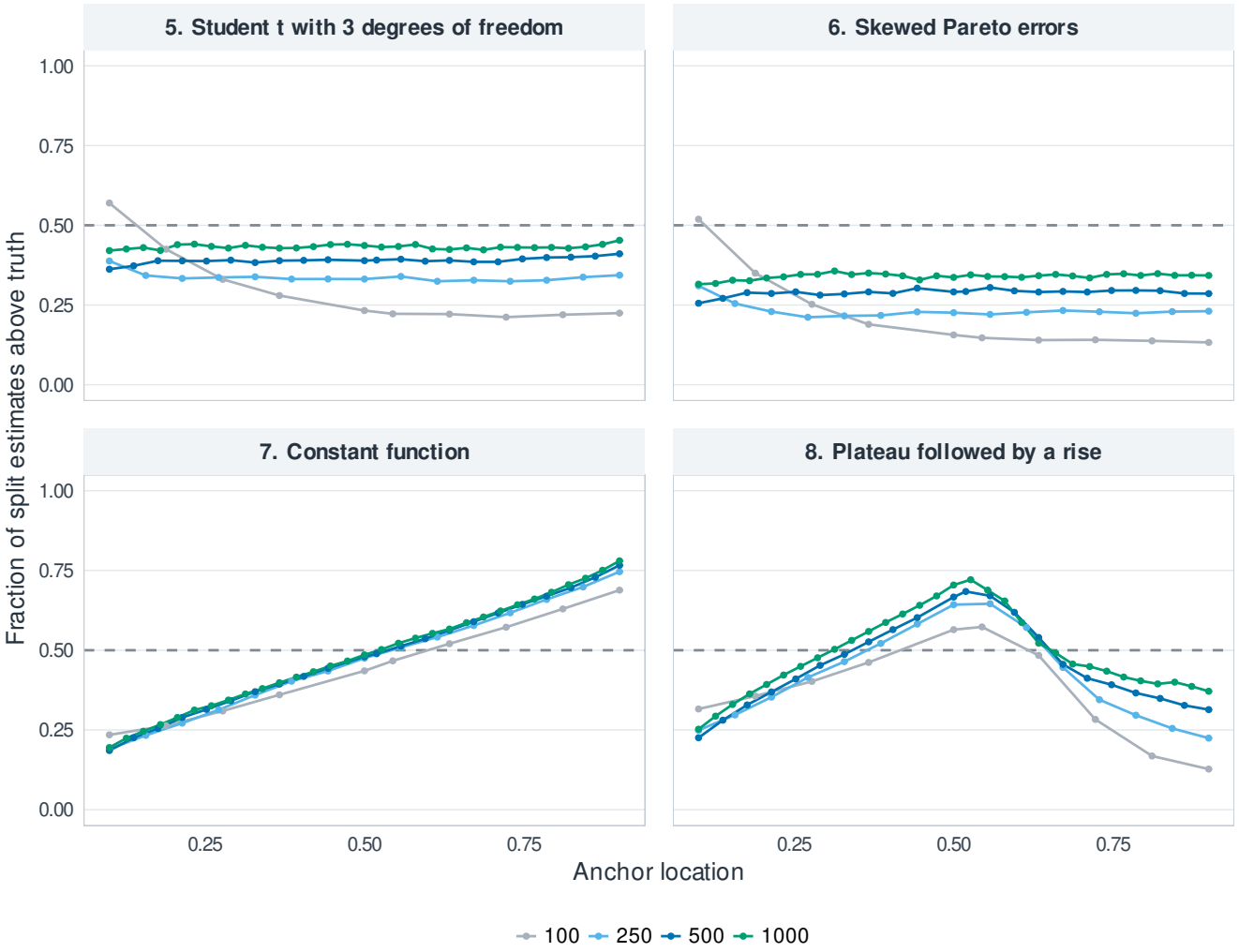


Figure 19: Fraction of the $9 \times 500 = 4500$ inference-block isotonic estimates above $f_0(t)$ at each anchor, for settings 5–8. Each curve corresponds to a total sample size, with half used for the pilot and half for the nine inference blocks. The dashed line marks one half. These finite-sample diagnostics show where the split estimates tend to lie above or below the target; they do not impose an asymptotic symmetry conclusion.

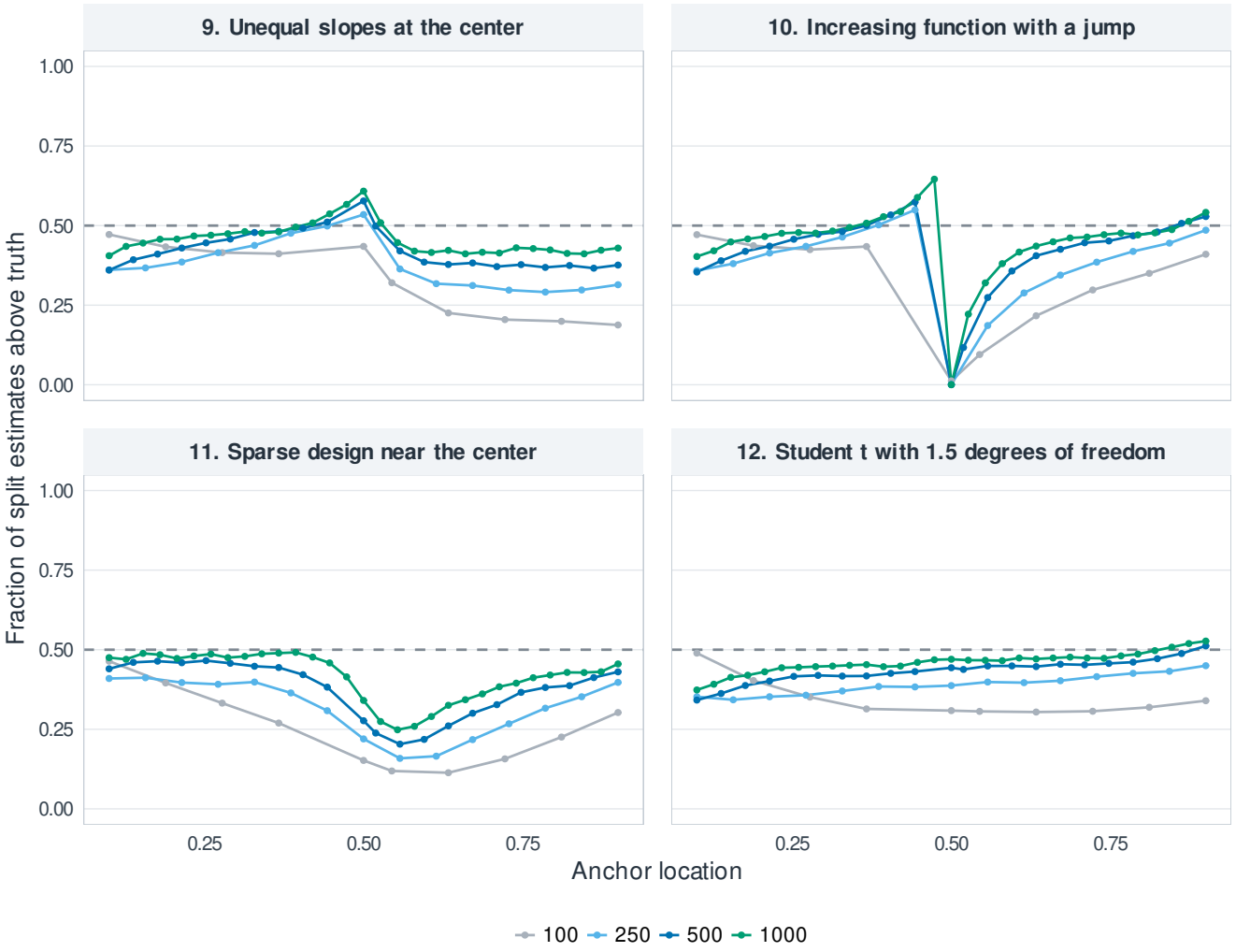


Figure 20: Fraction of the $9 \times 500 = 4500$ inference-block isotonic estimates above $f_0(t)$ at each anchor, for settings 9–12. Each curve corresponds to a total sample size, with half used for the pilot and half for the nine inference blocks. The dashed line marks one half. These finite-sample diagnostics show where the split estimates tend to lie above or below the target; they do not impose an asymptotic symmetry conclusion.

Reproducibility materials

The companion code and data reproduce the simulation, numerical summaries, and figures. The full simulation is invoked from the extracted code directory by

```
source("simulation_isotonic_example.R")
study <- run_isotonic_study(core_dir=".",
  output_dir="isotonic_results")
```

The script `plot_isotonic_saved_results.R` regenerates the figures from the recorded CSV files without rerunning the simulations. Computation supports forked workers on Linux and macOS and serial execution with `workers=1` on Windows. An optional C implementation accelerates isotonic fitting, with a base-R implementation available as a fallback. The results include package versions, random seeds, settings, and source hashes. The univariate calibration functions are taken without modification from COSI commit `3d4bcaa3f2099ae0348c481e1ba203315a78af8c`.

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